

Cryptology Assignment - 3

1) 1, 3, 5, 7, 9, 11, 15, 19, 21, 23, 25

$$3^{-1} \bmod 26$$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 + 2(-1) \quad \text{--- (1)}$$

$$2 = 26 + 3(-8) \quad \text{--- (2)}$$

$$1 = 3 + 2(-1)$$

$$1 = 3 + [26 + 3(-8)](-1)$$

$$1 = 3 + 26(-1) + 3(8)$$

$$3(9) + 26(-1) = 1$$

$$3^{-1} = 9$$

$$5^{-1} \bmod 26$$

$$26 = 5(5) + 1$$

$$26 + 5(-5) = 1$$

$$26 - 5 = 21$$

$$5^{-1} = 21$$

$$7^{-1} \bmod 26$$

$$26 = 7(3) + 5 \quad \rightarrow \quad 5 = 26 + 7(-3)$$

$$7 = 5(1) + 2$$

$$2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$5 + 2[7 + 5(-1)](-2) = 1$$

$$5 + 7(-2) + 5(2) = 1$$

$$26 + 7(-3) + 7(-2) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$7^{-1} = 15$$

9^{-1}

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$8 = 26 + 9(-2)$$

$$1 = 9 + [26 + 9(-2)](-1) = 1$$

$$9(3) + 26(-1) = 1$$

$$9^{-1} = 3$$

15^{-1}

$$26 = 15(1) + 11 \rightarrow 11 = 26 + 15(-1)$$

$$15 = 11(1) + 4 \rightarrow 4 = 15 + 11(-1)$$

$$~~11 = 26 + 15(-1)~~$$

$$11 = 4(2) + 3 \rightarrow 3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$15 + 11(-1) + (11 + 4(-2))(-1) = 1$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + (26 + 15(-1))(-2) + (15 + 11(-1))(2) = 1$$

11

$$15 + 26(-2) + 15(2) + 15(2) + 11(-2) = 1$$

$$15(5) + 26(-2) + 26 + 15(-1)(-2) = 1$$

$$15(5) + 26(-2) + 26(-2) + 15(2) = 1$$

$$15(7) + 26(-4) = 1$$

$$15^{-1} = 7$$

17⁻¹

$$26 = 17(1) + 9$$

$$9 = 26 - 17(1) = 9$$

$$9 = 8(1) + 1$$

$$9 = 8(-1) = 1$$

$$9 = 26 + 17(-1)$$

$$8 = 17 + 9(-1)$$

$$26 + 17(-1) + 17 + 9(-1)(-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-2) + 26 + 17(-1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26 - 3 = 23$$

$$17^{-1} = 23$$

19⁻¹

$$26 = 19(1) + 7 \rightarrow 7 = 26 + 19(-1)$$

$$19 = 7(2) + 5 \rightarrow 5 = 19 + 7(-2)$$

$$7 = 5(1) + 2 \rightarrow 2 = 7 + 5(-1)$$

$$7 = 26 + 19(-1)$$

$$19 = 5 - 2 \quad 5 + 2(-2) = 1$$

$$19 + 7(-2) + 7 + 5(-1)(-2) = 1$$

$$19 + 7(-2) + 7(-2) + 5(2) = 1$$

$$19 + 7(-4) + 5(2) = 1$$

$$+9 \neq 19 + (26 + 19(-1))(-4) + (19 + 7(-2))(2) = 1$$

$$19 + 26(-4) + 19(+4) + 19(2) + 7(-4) = 1$$

$$19(7) + 26(-4) + 26(-4) + 19(4) = 1$$

$$19(11) + 26(-8) = 1$$

$$19^{-1} = 11$$

$$21^{-1}$$

$$26 = 21(1) + 5 \rightarrow 5 = 26 + 21(-1)$$

$$21 = 5(4) + 1 \rightarrow 1 = 21 + 5(-4)$$

$$21 + 5(-4) = 1$$

$$21 + (26 + 21(-1))(-4) = 1$$

$$21 + 26(-4) + 21(4) = 1$$

$$26(-4) + 21(5) = 1$$

$$21^{-1} = 5$$

$$23^{-1}$$

$$26 = 23(1) + 3$$

$$3 = 26 + 23(-1)$$

$$23 = 3(7) + 2$$

$$2 = 23 + 3(-9)$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$26 + 23(-1) + (23 + 3(-7))(-1) = 1$$

$$26 + 23(-2) + 26(7) + 23(-7) = 1$$

$$26(8) + 23(-9) = 1$$

$$26 - 9 = 17$$

$$23^{-1} = 17$$

$$25^{-1}$$

$$26 = 25(1) + 1$$

$$26 + 25(-1) = 1$$

$$26 - 1 = 25$$

$$25^{-1} = 25$$

- 2) Choose any string of your choice to encrypt using the affine cipher with key value:
Multiplier 5 and additive 8

PLAIN TEXT : HARIS

$$\text{Formula : } C = (P \cdot \text{Multiplier} + \text{Additive}) \bmod 26$$
$$C = (P \times 5 + 8) \bmod 26$$

$$H = ((7 \times 5) + 8) \% 26 = 43 \% 26 = 17 = R$$

$$A = (0 \times 5 + 8) \bmod 26 = 8 \% 26 = 8 = I$$

$$R = (17 \times 5 + 8) \% 26 = 93 \% 26 = 15 = P$$

$$I = (8 \times 5 + 8) \% 26 = 48 \% 26 = 22 = W$$

$$S = (18 \times 5 + 8) \% 26 = 98 \% 26 = 18 = S$$

Cipher $\Rightarrow$ RIPWS

3) Encrypt 'HI' using affine cipher with $A=11$, $b=5$

$$C = (P \times a + b) \bmod 26$$

$$H=7$$

$$C = (P \times 11 + 5) \bmod 26$$

$$I=8$$

$$C = H = (7 \times 11 + 5) \% 26 = (77 + 5) \bmod 26 = 82 \bmod 26 = 4 = \underline{E}$$

$$I = (8 \times 11 + 5) \bmod 26 = 93 \% 26 = 15 = \underline{P}$$

Ciphertext = EP

4) The ciphertext UCR was encrypted using the affine function $(9x+2) \% 26$ find the plaintext.

$$U=20, C=2, R=17$$

$$C = (9P + 2) \% 26$$

$$P = a^{-1} (C - b) \% 26$$

$$a=9$$

$$b=2$$

$$9^{-1}$$

$$26 = 9(2) + 8$$

$$\underline{8} = 26 + 9(-2)$$

$$9 = 8(1) + 1$$

$$\underline{9} + 8(-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$9^{-1} = \underline{3}$$

$$U = 39(20-2) \bmod 26 = 54 \bmod 26 = 2 \Rightarrow C$$

$$C = 39(2-2) \bmod 26 = 0 \Rightarrow A$$

$$R = 39(17-2) \bmod 26 = 45 \bmod 26 = 19 \Rightarrow T$$

Plain text = CAT

5) Decrypt the ciphertext using affine cipher
 K D Y X J U W D E X Z Y using key 3 (multiplier) & 4

$$a=3 \quad b=4$$

$$P = a^{-1}(C-b) \bmod 26$$

K D Y X J U W D E X Z Y

10 3 24 23 9 20 22 5 4 23 25 24

$$a^{-1} = 3^{-1}$$

$$26 = 3(8) + 2$$

$$2 = 26 + 3(-8)$$

$$3 = (2)(1) + 1$$

$$3 + 2(-1) = 1$$

$$3 + 26 + 3(-8)(-1) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$26(-1) + 3(9) = 1$$

$$3^{-1} = 9 = a^{-1}$$

$$\begin{aligned}
 K &= 9(10-4) \% 26 = 9(6) \% 26 = 2 = C \\
 D &= 9(3-4) \% 26 = 9(-1) \% 26 = 17 = R \\
 Y &= 9(24-4) \% 26 = 9(20) \% 26 = 24 = Y \\
 X &= 9(23-4) \% 26 = 9(19) \% 26 = 15 = P \\
 J &= 9(9-4) \% 26 = 9(5) \% 26 = 19 = T \\
 U &= 9(20-4) \% 26 = 9(16) \% 26 = 14 = O \\
 W &= 9(22-4) \% 26 = 9(18) \% 26 = 6 = G \\
 D &= 9(3-4) \% 26 = 9(-1) \% 26 = 17 = R \\
 E &= 9(4-4) \% 26 = 9(0) \% 26 = 0 = A \\
 X &= 9(23-4) \% 26 = 9(19) \% 26 = 15 = P \\
 Z &= 9(25-4) \% 26 = 9(21) \% 26 = 7 = H \\
 Y &= 9(24-4) \% 26 = 9(20) \% 26 = 24 = Y
 \end{aligned}$$

Plaintext $\Rightarrow$ CRYPTOGRAPHY

- 6) Use $k_1 = 13$ (Multiplier), $k_2 = 4$ (Even though it is not a valid key $\rightarrow$ try to encrypt. Encrypt the string 'input' and 'alter' above key & find what is the issue.

$$C = (P \times \text{Multi} + \text{Add}) \% 26$$

$$C = (P \times a + b) \% 26$$

$$a = 13 \quad b = 4$$

13 is not co-prime of 26

Hence we can't find inverse of $13 \% 26$

Therefore, it is not a valid key for affine cipher.

For ~~top~~ 'input'

$$I = (8 \times 13 + 4) \% 26 = 108 \% 26 = 4 = E$$

$$N = (13 \times 13 + 4) \% 26 = 173 \% 26 = 17 = R$$

$$P = (15 \times 13 + 4) \% 26 = 199 \% 26 = 17 = R$$

$$U = (20 \times 13 + 4) \% 26 = 264 \% 26 = 4 = E$$

$$T = (19 \times 13 + 4) \% 26 = 251 \% 26 = 17 = R$$

For 'alter'

$$A = (0 \times 13 + 4) \bmod 26 = 4 \% 26 = 4 = E$$

$$L = (11 \times 13 + 4) \% 26 = 147 \% 26 = 17 = R$$

$$T = (19 \times 13 + 4) \% 26 = 251 \% 26 = 17 = R$$

$$E = (4 \times 13 + 4) \% 26 = 56 \% 26 = 4 = E$$

$$R = (17 \times 13 + 4) \% 26 = 225 \% 26 = 17 = R$$

Both PT will produce the same ciphertext.

The encryption is not one is one and decryption is not possible as 13 is not a valid key.

7) Find the inverse Matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 9 \end{bmatrix} \% 26$

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \% 26$$

$$\text{determinant}(K) = 3 \times 7 - 2 \times 5 = 21 - 10 = 11$$

$$11^{-1}$$

$$26 = 11(2) + 4$$

$$4 = 26 + 11(-2)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 + 3(1) = 1$$

$$26 + 11(-2) + 11 + 4(-2)(-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

Adj of mat

$$\begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$K^{-1} = 9 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \% 26$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \% 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8) Find the inverse of the following Mat whose entries are considered modulo 26:

$$\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \Rightarrow K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \pmod{26}$$

$$\det(K) = 11 \times 3 - 13 \times 2 = 33 - 26 = 7$$

$$\begin{aligned} 7^{-1} \quad & 26 = 7(3) + 5 \quad 5 = 26 + 7(-3) \\ & 7 = 5(1) + 2 \quad 2 = 7 + 5(-1) \\ & 5 = 2(2) + 1 \\ & 5 + 2(-2) = 1 \\ & 26 + 7(-3) + 2(7 + 5(-1))(-2) = 1 \\ & 26 + 7(-5) + 26(2) + 7(-6) = 1 \\ & 26(3) + 7(-11) = 1 \\ & 26 - 11 = 15 \end{aligned}$$

$$\text{Adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9) Is the key valid?

$$\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$$

For a key to be valid, its determinant should be co-prime & shouldn't be equal to 0.

$$\det(K) = 2(21) - 14(13)$$
$$= 42 - 182$$

$$= -140$$

$$-140 \bmod 26 = 16$$

$$\det(K) = 16 \bmod 26$$

16 is not coprime to 26.

Therefore it will not have an inverse $\% 26$

therefore the key $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$ is not valid.

10) Encrypt using 2×2 Hill cipher for the PT

'SOLVED' Using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

$$3 \times 1 - 7 \times 8 = -23 = 3 \quad (\text{co-prime})$$

$$C = K \times P$$

So | L V | E D

$$\begin{matrix} S \\ O \end{matrix} \cdot P = \begin{bmatrix} 18 \\ 14 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 54 + 112 \\ 98 + 154 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \% 26$$

$$166 \bmod 26 = 10 = K$$

$$280 \bmod 26 = 20 = V$$

$$\begin{matrix} L \\ V \end{matrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 33 + 168 \\ 77 + 231 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \% 26$$

$$201 \bmod 26 = 19 = T$$

$$308 \% 26 = 22 = W$$

$$\begin{matrix} E \\ D \end{matrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 12 + 24 \\ 28 + 33 \end{bmatrix} \% 26 = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \% 26$$

$$36 \% 26 = 10 = K$$

$$61 \% 26 = 9 = J$$

SOLVED $\rightarrow$ K U T W K J